

AWS Cloud Practical

Name: Pooja Prakash Belgaumkar

1. RDS MySQL and EC2 connection setup

- Create a MySQL relational database using the Standard Create method in Amazon RDS.
- Provided the master username as admin and password as admin123 for database authentication.
- Selected the default VPC, which contains subnets across multiple Availability Zones.

The screenshot displays the AWS Management Console interface for an Amazon RDS database instance. The top navigation bar shows the AWS logo, a search bar, and the user's profile (PoojaPrakash (7871-6932-0097)). The breadcrumb trail indicates the path: Aurora and RDS > Databases > database-2. The instance name 'database-2' is prominently displayed at the top of the console, along with 'Modify' and 'Actions' buttons. Below this, a 'Summary' section provides key details: DB identifier (database-2), Status (Available), Role (Instance), Engine (MySQL Community), CPU (6.29%), Class (db.t3.micro), Current activity (0 Connections), and Region & AZ (eu-north-1a). A 'Recommendations' section is also visible. The 'Connectivity & security' tab is selected, showing three connection methods: 'Code snippets' (selected), 'CloudShell', and 'Endpoints'. Below this, the 'Internet access gateway' is set to 'Disabled' and 'IAM Authentication' is also 'Disabled'.

aws | Europe (Stockholm) | PoojaPrakash (7871-6932-0097) | PoojaPrakash

Aurora and RDS > Databases > database-2

database-2

Modify Actions

Summary

DB identifier database-2	Status Available	Role Instance	Engine MySQL Community
CPU 6.29%	Class db.t3.micro	Current activity 0 Connections	Region & AZ eu-north-1a

Recommendations

< Connectivity & security | Monitoring | Logs & events | Configuration | Zero-ETL integration >

Connect using Info

☒ **Code snippets**
Use when connecting through SDK, APIs, or third-party tools including agents.

☐ **CloudShell**
Use for a quick access to AWS CLI that launches directly from the AWS Management Console.

☐ **Endpoints**
Use when connecting through any IDE interface.

Internet access gateway
Disabled

IAM Authentication
Disabled

Programming language

MySQL (macOS) ▼

Endpoint type

Instance endpoint ▼

Connection steps

Follow the steps below to paste the code of each step in your tool and run the commands. The snippets dynamically reflect the authentication configuration.

```
1 mysql -h database-2.cb0i842a4kzs.eu-north-1.rds.amazonaws.com -P 3306 -u admin -
  p'<Enter_DB_Password>' --ssl-verify-server-cert --ssl-ca=/certs/global-bundle.pem mysql
```



Connected compute resources (0) Info



Actions ▼

Connections to compute resources that were created automatically by RDS are shown here. Connections to compute resources that were created manually aren't shown.

Filter by compute resources

< 1 > ⚙️

Resource identifier ↗ ▲ | Resource type ▼ | Availability Zone ▼ | VPC security group ↗ ▼ | Compute resource s

No connected compute resources

No connected compute resources that were created automatically to display.

Set up EC2 connection

Set up Lambda connection

Security group rules (1)



Filter by Security group rules

< 1 > ⚙️

Security group ▲	Type ▼	Rule ▼
launch-wizard-31 (sg-04aef27e552f17292)	CIDR/IP - Outbound	0.0.0.0/0

Replication (1)



Filter by Replication

< 1 > ⚙️

DB identifier ▲	Role ▼	Region & AZ ▼	Replication source ▼	Replication state ▼	Lag ▼
database-2	Instance	eu-north-1a	-	-	-

- Launch one EC2 instance (Ubuntu) to act as the client machine for connecting to the RDS database.

EC2 > Instances > i-0323c0d80a7fe5f7f

Instance summary for i-0323c0d80a7fe5f7f (EC2 Instance) [Info](#)

[Refresh](#) [Connect](#) [Instance state ▼](#) [Actions ▼](#)

Updated less than a minute ago

Instance ID i-0323c0d80a7fe5f7f	Public IPv4 address 51.20.128.45 open address ↗	Private IPv4 addresses 172.31.6.51
IPv6 address -	Instance state Running	Public DNS ec2-51-20-128-45.eu-north-1.compute.amazonaws.com open address ↗
Hostname type IP name: ip-172-31-6-51.eu-north-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-6-51.eu-north-1.compute.internal	
Answer private resource DNS name -	Instance type t3.micro	Elastic IP addresses -
Auto-assigned IP address 51.20.128.45 [Public IP]	VPC ID vpc-04fbb5dfae5f66a87	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more ↗
IAM role -	Subnet ID subnet-0cbc6ee27b50c1eda (subnet1)	Auto Scaling Group name -

- Configured connectivity by editing the inbound rules of the RDS security group to allow the EC2 instance to communicate with the database. Added a rule with:
 - Type: MySQL/Aurora
 - Port: 3306
 - Source: EC2 instance's security group

EC2 > Security Groups > sg-04aef27e552f17292 - launch-wizard-31 > Edit inbound rules

Description - optional [Info](#)

Inbound rule 3 [Delete](#)

Security group rule ID: -

Type: [Info](#) MySQL/Aurora

Protocol: [Info](#) TCP

Port range: [Info](#) 3306

Source type: [Info](#) Custom

Source: [Info](#) sg-08c36d14cb03b7936
sg-08c36d14cb03b7936

Description - optional [Info](#)

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. [X](#)

[Cancel](#) [Preview changes](#) [Save rules](#)

- Connect to the EC2 instance and installed the MySQL client package.

```
ubuntu@ip-172-31-6-51:~$ mysql --version
mysql Ver 8.0.45-0ubuntu0.24.04.1 for Linux on x86_64 ((Ubuntu))
ubuntu@ip-172-31-6-51:~$
```

- Established a connection to the RDS database from the EC2 instance using the RDS endpoint and authenticated using the configured database credentials.

```
ubuntu@ip-172-31-6-51:~$ mysql -h database-2.cb0i842a4kzs.eu-north-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 35
Server version: 8.4.7 Source distribution

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

- Verified the database connection by executing the *SHOW DATABASES;* command. Additionally, created a new database using the *CREATE DATABASE <database_name>;* command.

```
mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.02 sec)

mysql> CREATE DATABASE pooja_test;
Query OK, 1 row affected (0.01 sec)

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| pooja_test |
| sys |
+-----+
5 rows in set (0.00 sec)

mysql> exit;
Bye
```

- This task demonstrates the successful setup of an Amazon RDS MySQL database and secure connectivity from an EC2 instance using security group configuration and endpoint-based authentication. The implementation validates a real-world cloud architecture where application servers communicate securely with managed database services within a VPC environment.